# ENGINEERING: DEVELOPMENT AS A PROFESSIONAL & ROUTES TO REGISTRATION

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# Engineers…

- An engineer is a professional practitioner, concerned with applying scientific knowledge, mathematics, to develop solutions for technical problems.
- Engineers design materials, structures, and systems while considering the limitations imposed by practicality, regulation, safety, and cost.
- The word engineer is derived from the Latin roots ingeniare ("to contrive, devise") and ingenium ("cleverness").
- Engineers forms the link between scientific discoveries and their applications to human needs and quality of life.

# Note: Title '*engineer*' not protected by UK law

- therefore anyone can call themselves an *engineer* or *professional engineer* or *registered engineer*
- many semi-skilled and unskilled trades adopt this title

# A bit of recent history…

- Starting with the Industrial Revolution 1750-1850
  - Rural working the land >>>>urban manufacture & commerce
  - Development of canals networks, locks and aqueducts, new docks in London, Bristol, Liverpool and Newcastle…
  - A vast increase in City and Town populations e.g. between 1800 to 1850 London expanded from 1 million to 2.5 million.
- Which drove technological development….

# Quiz…

1. Who developed the first public intercity railway and when?
2. What was patented in 1876 by Alexander?
3. Who developed the first working four stroke engine?
4. The first UK motorway was opened in …. around which town?
5. The first widely available plastic was called …. and its primary application was…?
6. In which country was the first civilian nuclear power plant built?
7. Smeaton was most famous for which type of structure?
8. Which three engineers developed the transistor?
9. James Martin developed what and when?
10. Isambard Kingdom Brunel is famous for which bridge?

# 1: First intercity railway 1830

4 feet 8 1/2 inches...

Stevenson gauge

What else is he famous for?

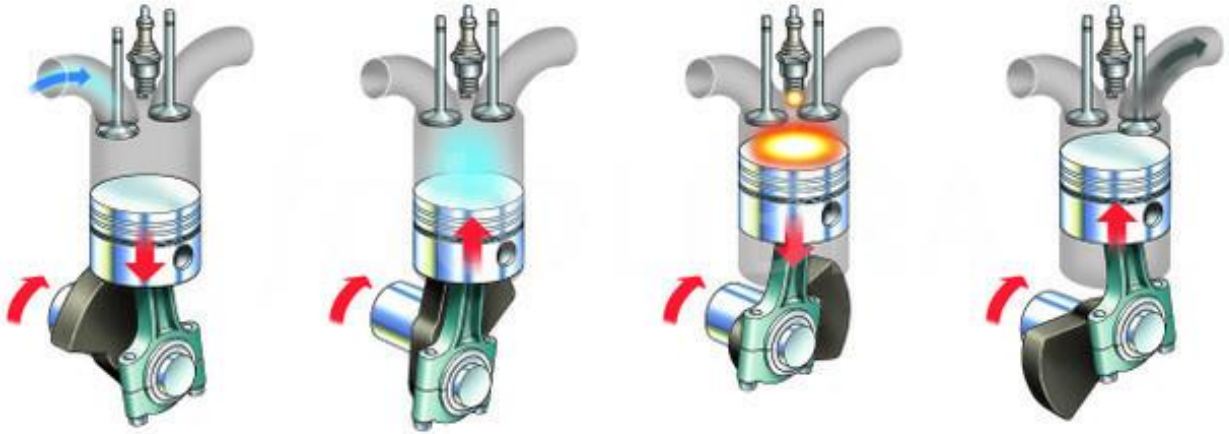
# 2: Telephone 1876

Alexander Graham Bell

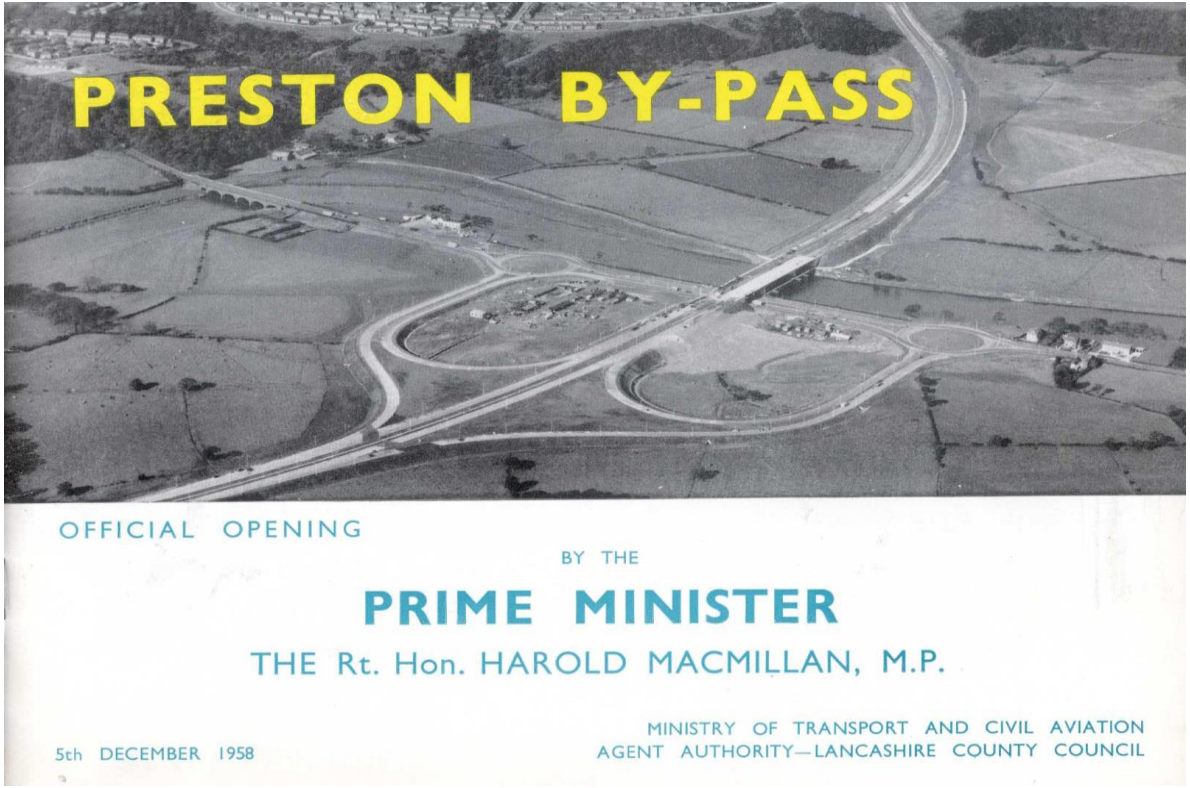
To this....

# 3: Nikolaus Otto 1876

First 'effective' working four stroke engine



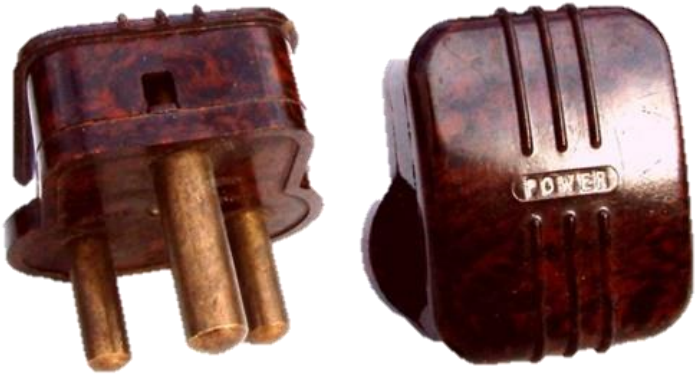
# 4: First Motorway 1958



# 5: Bakerlite 1907

- Developed by Belgian chemist Leo Baekeland in New York

- Used in emerging electrical and automotive industries due to its extremely high resistance to electricity, heat and chemical action



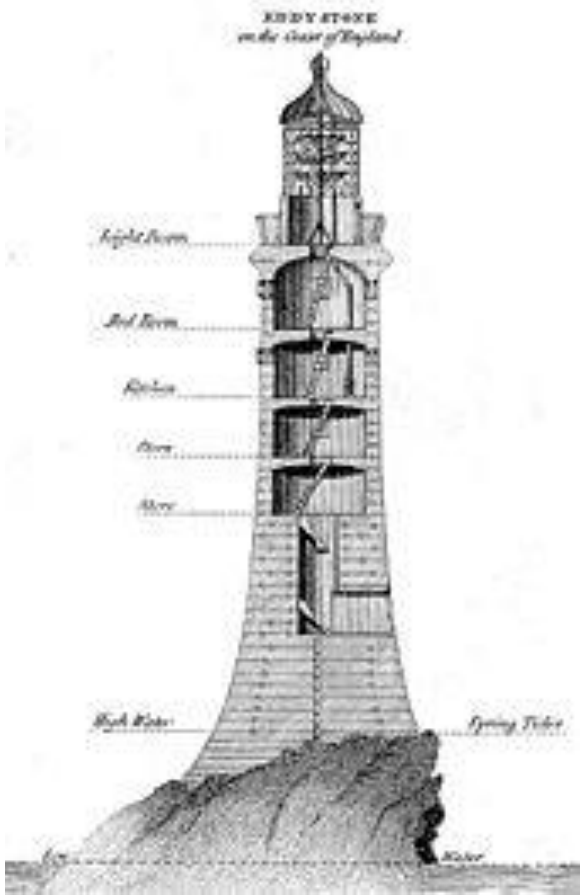
# 6: Russia 1954

- Obninsk Nuclear Power Station - Decommissioned 1994

# 7: Lighthouse completed 1759

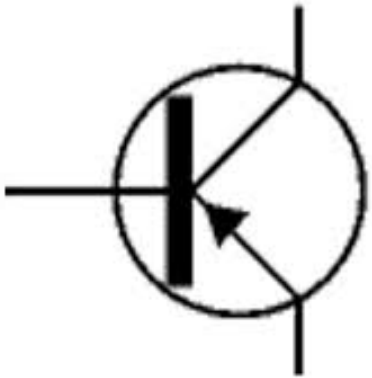
- First to sign himself ‘civil engineer’ to distinguish from the military engineers.
- Developed hydraulic lime mortar to set underwater and pioneered the use of Dove-tailed joints

*Eddystone lighthouse*



# 8: Transistor 1948

- John Bardeen, William Shockley and Walter Brattain



Transistor

# 9: Ejection seat - 1944

- 'Martin's Aircraft Works' was founded by Sir James Martin in 1934.
- On 12th September 1942, during a test flight, Captain Valentine Baker was killed greatly affected Sir James Martin, and making pilot safety his primary focus.
- After investigating alternatives, the most attractive means was forced ejection of the seat with the occupant sitting in it - the ejection seat.

# 10: Clifton Suspension bridge

# A bit more history…

- In the UK, the importance of engineers as soldiers was recognised by the founding of the Corps of Engineers in 1717.
- Recognition of the contribution to civilian society came 100 years later when the Institution of Civil Engineers was formed in 1818.
- Civil Engineering > industrial revolution > particularly canals, bridges, lighthouses, ports and public health.
- The significance of the railways led to creation of the Institution of Mechanical Engineers in 1847
- Technological advancements in the use of electricity in communications led to the establishment of the Institution of Electrical Engineers in 1871.
- As engineering became more specialised, the number of institutions grew.

# Engineering Institutions

| | | |
|---|---|---|
| **IMechE** | ICME | IOP |
| **IET** | IED | IPEM |
| **ICE** | IFE | IRSE |
| BCS | IGEM | IStructE |
| BINDT | IHE | IoW |
| CIBSE | IHEEM | NI |
| CIHT | ILP | RAeS |
| CIPHE | IMarEST | RINA |
| CIWEM | InstMC | SEE |
| EI | InstRE | SOE |
| IAgrE | IOA | TWI |
| IChemE | IOM3 | |

***80% of 250,000 professionally registered engineers are members of The 'Big Three' .***

# Regulation

- By the mid 1950s there was an obvious need for a central body to agree standards and represent the engineering profession

- This led to the creation of the :

  **Joint Council of Engineering Institutions**

  *which became*

  **The Council of Engineering Institutions**,

  *before becoming the…*

  **Engineering Council EC$^{UK}$**

  Competence outlined in *UK-SPEC*

# The role of EC$^{UK}$

- Britain's regulatory authority for registration of Chartered and Incorporated Engineers and Engineering Technicians
- Set the standards for engineering
- responsible for the accreditation of educational and training programmes through Licensed Member Institutions.

# Professional qualification

- EC$^{UK}$ insist the process is robust
- It takes a number of years to complete.
- A combination of formal education and experience generally known as Initial Professional Development (IPD).
- During IPD, experience is regularly reviewed (usually on a quarterly basis) by a professional engineer acting as a Mentor.
- Only when an individual has demonstrated that they have met all of the competencies should they go forward for professional review.
  - Application and Interview

# A life long commitment

- Graduates
  - Quarterly reports
  - Annual Assessments
- All Applicants
  - Final Application
  - Rigorous review
  - Ongoing Auditable CPD

- Some of this will be in your own time

# IPD Routes to registration

**Undergraduates**

Eng Tech using Year in Industry as experience

*or*

IMechE MPDS

IET Career Manager

IoP MyCareerPath

**Graduates**

- IMechE MPDS
- IET Career Manager
- IoP MyCareerPath
- Other PEI Recognised IPD reporting sytems

# Direct Routes to registration

- Application via Appropriate PEI
  - IMechE
    - Register on IMechE website
      - Career Learning Appraisal may be required if no degree or not recognised
      - 2 stage process before direct application
    - Complete Application
  - IET
    - Join IET (Fee) and apply
      - Be ready to provide additional information if no or un-recognised degree

# Importance to JLR

- The JLR name is synonymous worldwide with world class quality: all our vehicles are designed and built for hard work and reliability
- The JLR name should also be synonymous worldwide with world class engineers
  - JLR want to support that goal
  - Develop more rounded engineers for the business
  - Identifies talent for progression and succession

# Competence

*Competence is the ability to carry out a task to an effective standard. To achieve competence requires the right level of knowledge, understanding and skill, and a professional attitude.*

Experience

## Competence

Understanding

Values

Further learning

Skills

Knowledge

Education

*Competence is developed by a combination of formal and informal learning, training and experience. Competences set out in UK SPEC EC[UK]*

# UK-SPEC – 5 Competences

A. Knowledge and understanding
B. Design and development of processes, systems, services and products
C. Responsibility, management or leadership
D. Communication and inter-personal skills
E. Professional commitment

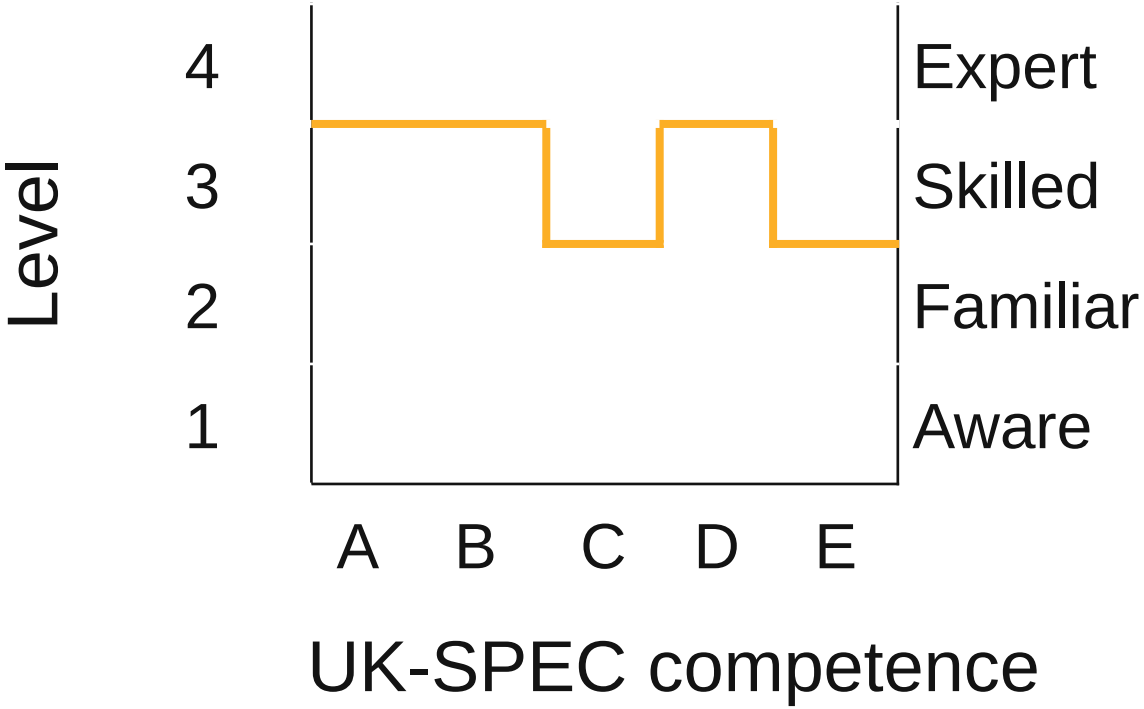
# Competence levels

| Level | Descriptor |
|---|---|
| 1<br>Aware | • Performs the activity with significant supervision and guidance<br>• Performs basic routine and predictable tasks<br>• Little or no individual responsibility |
| 2<br>Familiar | • Performs the activity in a range of contexts<br>• Supervision only necessary in more complex circumstances<br>• Some individual responsibility or autonomy |
| 3<br>Skilled | • Performs activity in some complex and non-routine contexts<br>• Significant responsibility and autonomy<br>• Can oversee others' work |
| 4<br>Expert | • Performs the activity in a wide range of complex and non-routine contexts<br>• Substantial personal autonomy<br>• Can develop others in the activity |

# Competence vs. Level

*Min requirement at professional review*

*3 x Level 3 and 2 x Level 2*



# Recording Competence

Needs to go beyond the recording of facts

Experience is not only seeing or doing – it is asking what you learnt from it.

Important to ‘tell a tale’

Suggest using Kipling’s (1902)

- ‘Honest serving men’

I Keep six honest serving-men,
they taught me all I knew;
Their names are **What** and **Why** and **When**,
and **How** and **Where** and **Who**.

# Recording Competence

## Tell a tale!!!

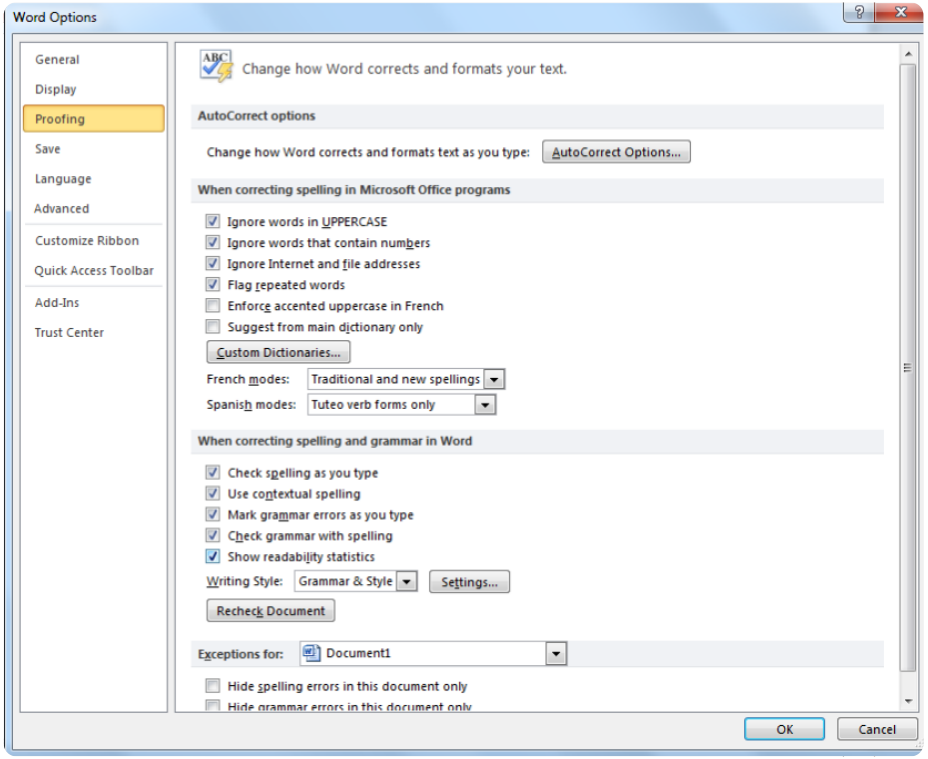
**25% Context**

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

**75% Your contribution/learning**

# Switching on Readability Statistics in MS Word

1. Click the **Microsoft Office Button** , and then click **Word Options**.
2. Click **Proofing**.
3. Make sure **Check grammar with spelling** is selected.
4. Under **When correcting grammar in Word**, select the **Show readability statistics** check box.



# Understanding readability scores

## Flesch Reading Ease test

This test rates text on a 100-point scale. The higher the score, the easier it is to understand the document. It is recommended that for most standard files, you want the score to be between 60 and 70. Unlikely in engineering as we use complicated words – ones with lots of syllables.

***For the geeks amongst you***

$$\text{Flesch Reading Ease} = 206.835 - (1.015 \times ASL) - (84.6 \times ASW)$$

where:

ASL = average sentence length (the number of words divided by the number of sentences)

ASW = average number of syllables per word (the number of syllables divided by the number of words)

**Shorter sentences are better**

**Don’t use complicated words if you don’t need to**

# Understanding readability scores

## Flesch-Kincaid Grade Level test

This test rates text on a U.S. school grade level. For example, a score of 8.0 means that an eighth grader can understand the document.

Flesch-Kincaid Grade Level = (.39 x ASL) + (11.8 x ASW) – 15.59

where:

ASL = average sentence length (the number of words divided by the number of sentences)

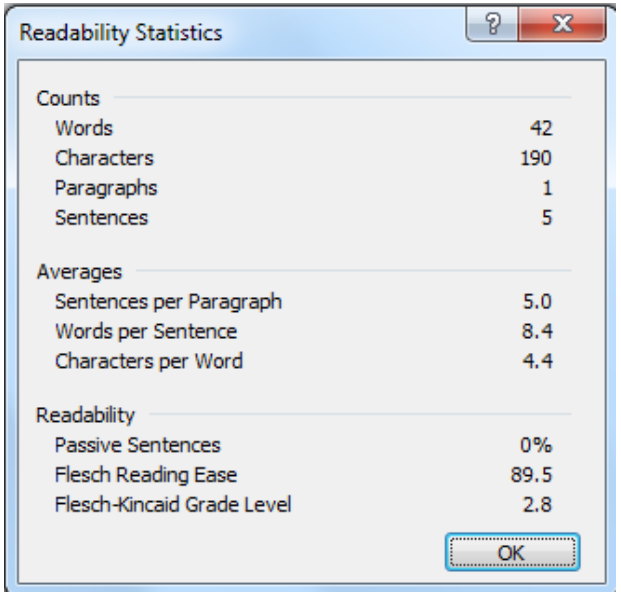
ASW = average number of syllables per word (the number of syllables divided by the number of words)

**The lower the grade level the better.**

# The Three Billy Goats Gruff

Once upon a time there were three goats. They were the Billy Goats Gruff. There was a little billy goat with little horns. There was a middle-sized billy goat with middle-sized horns. There was a big billy goat with very big horns.

© M&S First Readers



Readability Statistics

| | |
|---|---|
| **Counts** | |
| Words | 42 |
| Characters | 190 |
| Paragraphs | 1 |
| Sentences | 5 |
| **Averages** | |
| Sentences per Paragraph | 5.0 |
| Words per Sentence | 8.4 |
| Characters per Word | 4.4 |
| **Readability** | |
| Passive Sentences | 0% |
| Flesch Reading Ease | 89.5 |
| Flesch-Kincaid Grade Level | 2.8 |

OK

# Passive sentences

In a passive sentence, the subject does not perform the action in the sentence, the action is performed on it. For example:

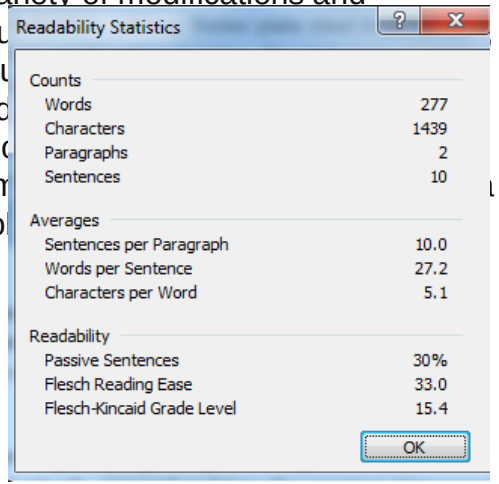
Anita was driven to the theatre.

In this example, Anita did not perform the action of the verb to drive. She was the recipient of the action.

# Extract from a quarterly report

The first of these projects involved analysing a new hydraulic excavator body structure that was under development so that I could offer the design team some new and fresh ideas on how to improve its structural rigidity. The issue with the body was that when it was placed under load it triggered excessive bending and twisting moments therefore demonstrating that the structure was too flexible.

Upon beginning this analysis I firstly organised numerous discussions with members of the design team about the problems they were encountering. On receipt of these discussions I obtained evidence in the form of finite element analysis models showing the areas of high stress concentration. From these I was able to conduct an in depth evaluation of the entire structure using my previous practical and theoretical knowledge. Over a period of time, I managed to produce a variety of modifications and additions to the structure that could be applied. These included the addition of strengthening gu utilising thicker plate steel in certain areas and incorporating a baffle design in the box style stru The main issue I encountered when developing these additions were such that they had limited implemented due to service access and component placement. I overcome this issue by develo bracing fixture that could be removed at service intervals to give access to the surrounding com position with sufficient improvements I organised a meeting comprising of my manger and a co portray my ideas across.



# Extract from a quarterly report

Does this sentence add value?

The first of these projects involved analysing a new hydraulic excavator body structure that was under development so that I could offer the design team some new and fresh ideas on how to improve its structural rigidity. The issue with the body was that when it was placed under load it triggered excessive bending and twisting moments therefore demonstrating that the structure was too flexible.

One or the other.

Did you do this…it's unclear?

Upon beginning this analysis I firstly organised numerous discussions with members of the design team about the problems they were encountering. On receipt of these discussions I obtained evidence in the form of finite element analysis models showing the areas of high stress concentration. From these I was able to conduct an in depth evaluation of the entire structure using my previous practical and theoretical knowledge. Over a period of time, I managed to produce a variety of modifications and additions to the structure that could be applied. These included the addition of strengthening gussets in places of most deflection, utilising thicker plate steel in certain areas and incorporating a baffle design in the box style structure that would reduce twisting. The main issue I encountered when developing these additions were such that they had limited space in which they could be implemented due to service access and component placement. I overcome this issue by developing the idea of a non-permanent bracing fixture that could be removed at service intervals to give access to the surrounding components. Once I found myself in a position with sufficient improvements I organised a meeting comprising of my manger and a collection of the design engineers to portray my ideas across.

FEA is strain based.

State where developed

Numbers are good! What deflection is acceptable?

This is good although it should be a separate paragraph could be more succinct. What was the outcome?

# This was my attempt at the last paragraph

The main issues I encountered when developing the additions were packaging constraints due to service access and component placement. I overcome this by developing a non-permanent bracing fixture that could be removed at service intervals to give access. To portray my ideas, I organised a meeting with my manger and the design engineers.

# Overall comparison using statistics

The first of these projects involved analysing a new hydraulic excavator body so that I could offer the design team some ideas to improve its structural rigidity. To start the analysis, I liaised with the design team to identify the problems they were encountering. I then reviewed their finite element analysis models to identify areas of concern. I then conducted an in depth evaluation of the entire structure using my previous practical and theoretical knowledge.

In a few weeks, I designed a variety of structural modifications and additions. These included the addition of strengthening gussets where deflection
other areas and incorporating a baffle in the box st

The main issue I encountered when developing the
service access and component placement. I overc
bracing fixture that could be removed at service int
organised a meeting with my manger and the desi

**Before modification**

| | |
|---|---|
| **Counts** | |
| Words | 277 |
| Characters | 1439 |
| Paragraphs | 2 |
| Sentences | 10 |
| **Averages** | |
| Sentences per Paragraph | 10.0 |
| Words per Sentence | 27.2 |
| Characters per Word | 5.1 |
| **Readability** | |
| Passive Sentences | 30% |
| Flesch Reading Ease | 33.0 |
| Flesch-Kincaid Grade Level | 15.4 |

OK

**This version**

| | |
|---|---|
| **Counts** | |
| Words | 171 |
| Characters | 918 |
| Paragraphs | 1 |
| Sentences | 9 |
| **Averages** | |
| Sentences per Paragraph | 9.0 |
| Words per Sentence | 19.0 |
| Characters per Word | 5.2 |
| **Readability** | |
| Passive Sentences | 11% |
| Flesch Reading Ease | 34.6 |
| Flesch-Kincaid Grade Level | 13.1 |

OK

# Competence Levels

I was told to conduct an evaluation of the structure using my previous practical and theoretical knowledge…

*is very different from*

I decided to conduct an in depth evaluation of the entire structure using my previous practical and theoretical knowledge…

Remember you need to be demonstrating:

**Level 3** Performs the activity in some complex and non-routine contexts; significant responsibility and autonomy; can oversee the work of others.

# The next steps are down to you!

It’s got to be something you want!

You need to:

- Consider if this is right for you and you are willing to commit the time and effort
- I will put your name on a master list if it is
- You then get on with your application with help from:
  - Me
  - Your Colleagues
  - Registered Engineers in your area